Dr. A. Q. Khan Institute of Computer Sciences and Information Technology
KICSIT, IST Kahuta Campus | Fall 2026

OPERATING SYSTEM
ASSIGNMENT NO.1

SUBMITTED BY:

NAME: IMMAN FATIMA

CLASS: BSCS 5'A'

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SUBMITTED TO:

SIR UZAIR HASSAN

SUBMISSION DATE:

02 NOV 2026

Title: Windows System Administration & Task Automation Challenge

Scenario

You have recently joined an organization as a Junior System Administrator. Your organization has provided you with a new Windows computer that needs to be organized, configured, and partially automated.

Your task is to use Windows features, keyboard shortcuts, Command Prompt (CMD), Batch Scripting, and PowerShell to perform the required system-administration activities.

You must practically perform every task on your own computer and prepare a Lab Assignment Report containing screenshots, commands/scripts, and a short explanation of your work.

Task 1 — Windows & Keyboard Shortcuts

Demonstrate at least **10 useful Windows keyboard shortcuts** that can improve the productivity of a system administrator.

For each shortcut:

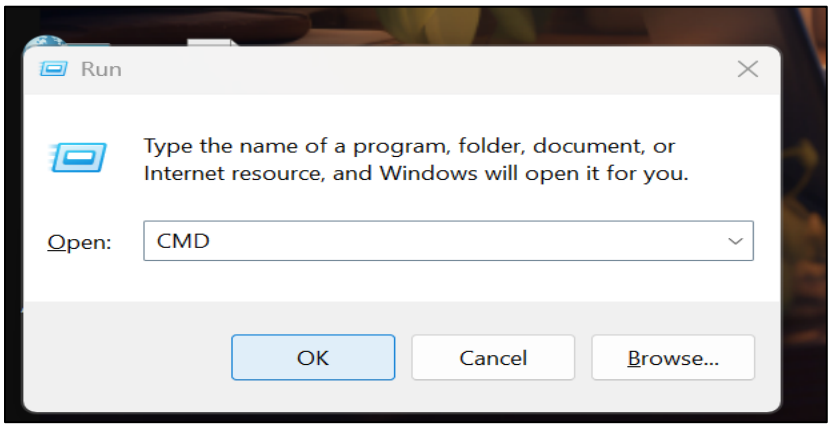
- Write the shortcut keys.
- Explain its purpose in one sentence.
- Practically demonstrate it.
- Add a screenshot where appropriate.

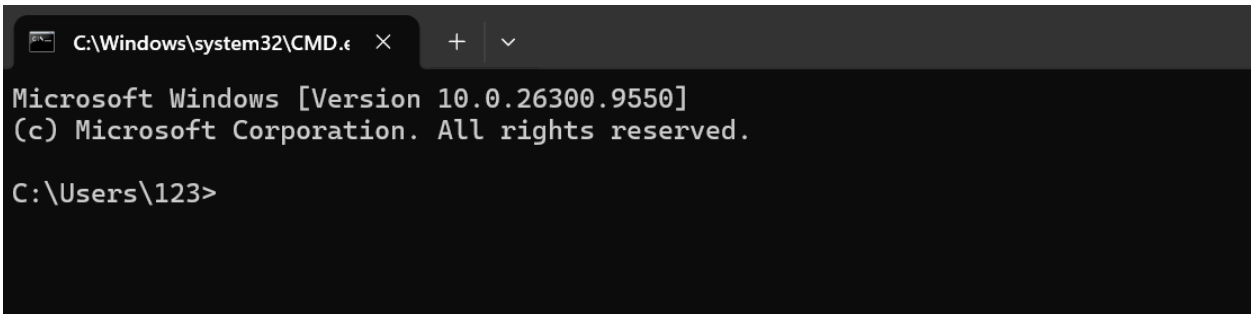
At least 5 shortcuts must involve Windows-key combinations.

- **Win + R**

PURPOSE:

Opens the Run box to launch tools such as cmd or powershell quickly.

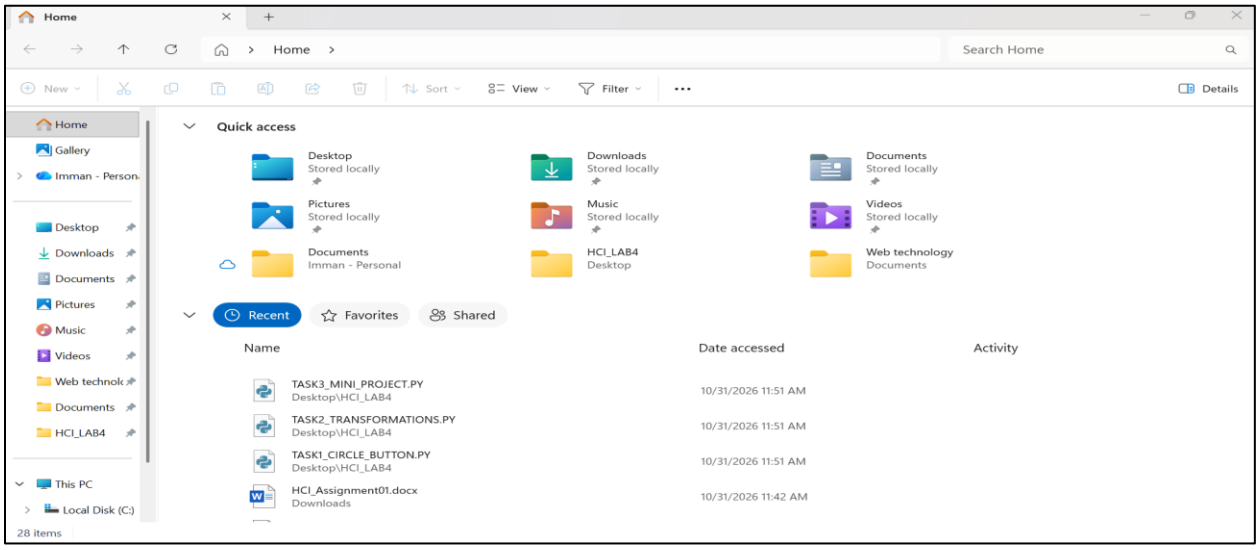




- **Win + E**

PURPOSE:

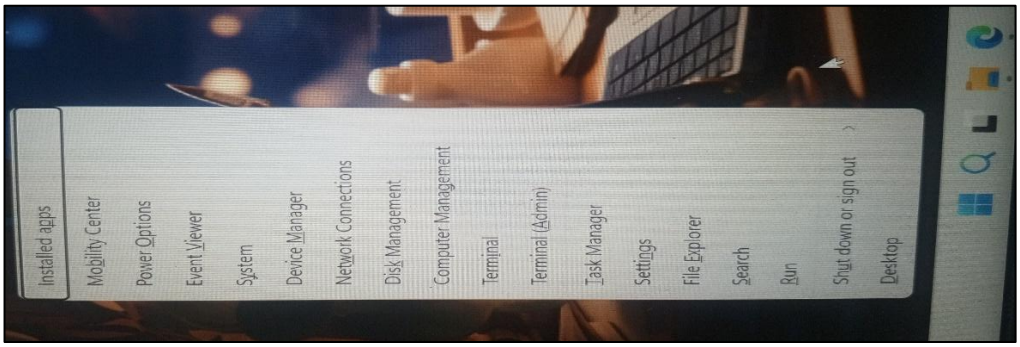
Opens File Explorer.



- **Win + X**

PURPOSE:

Opens the power-user menu (Task Manager, Terminal, Device Manager).



- **Win + L**

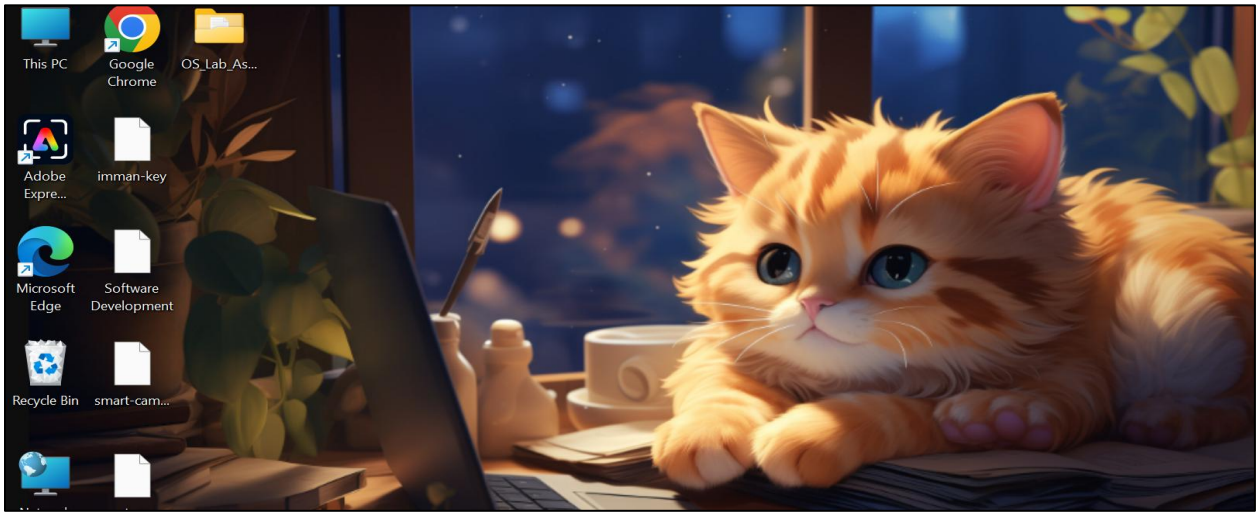
PURPOSE:

Locks the computer to keep it secure.

- **Win + D**

PURPOSE:

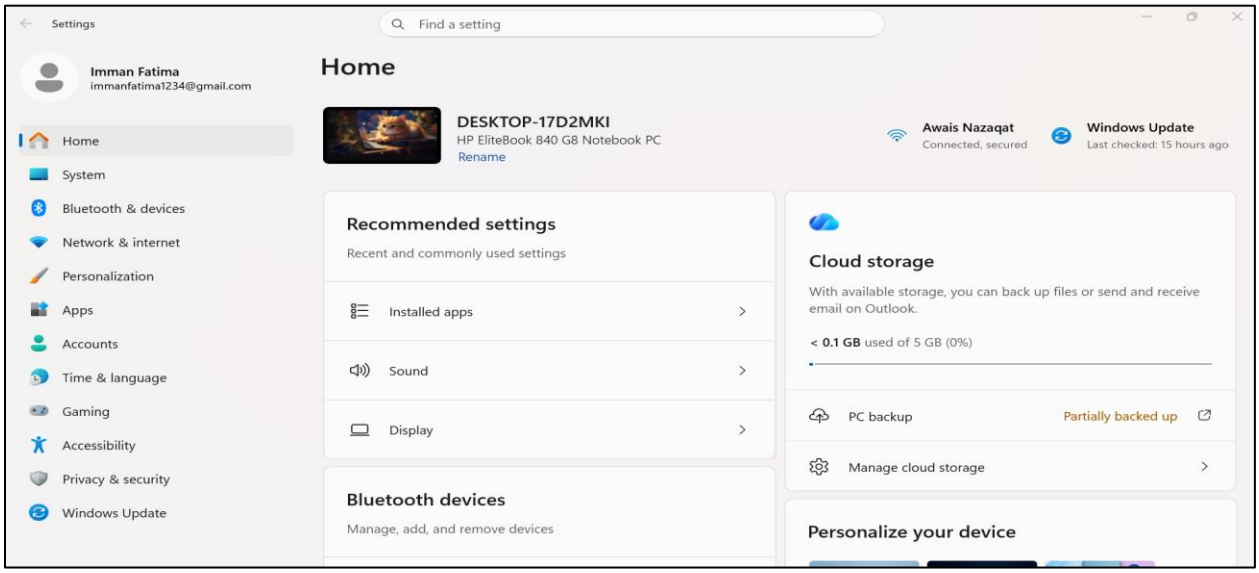
Shows or hides the desktop.



- **Win + I**

PURPOSE:

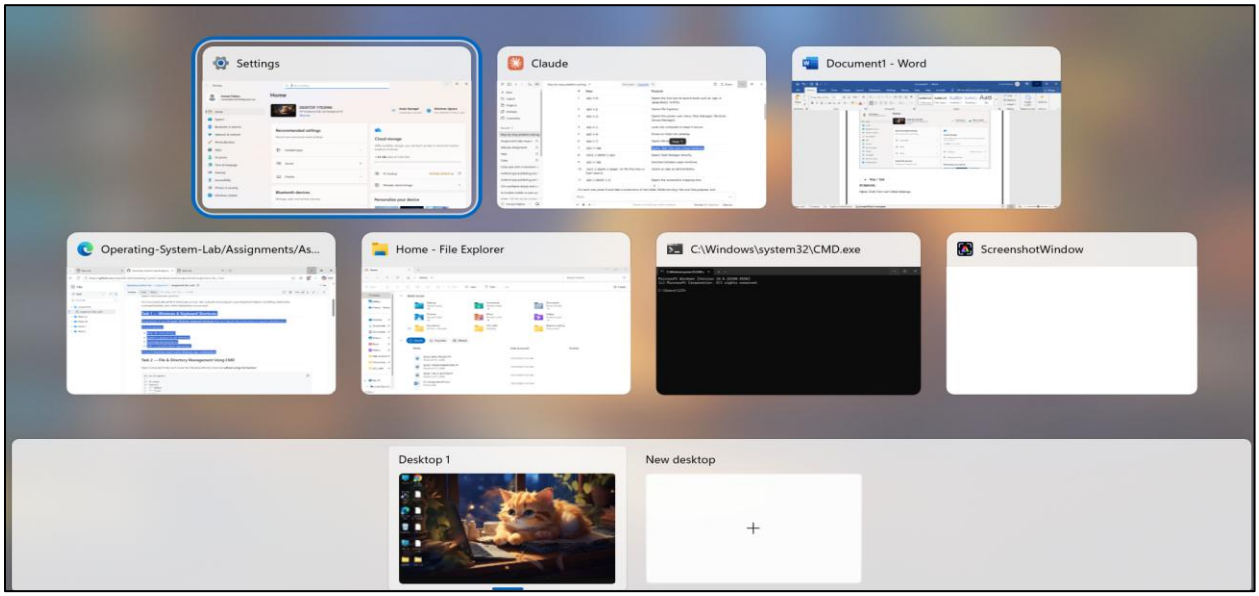
Opens Windows Settings.



- **Win + Tab**

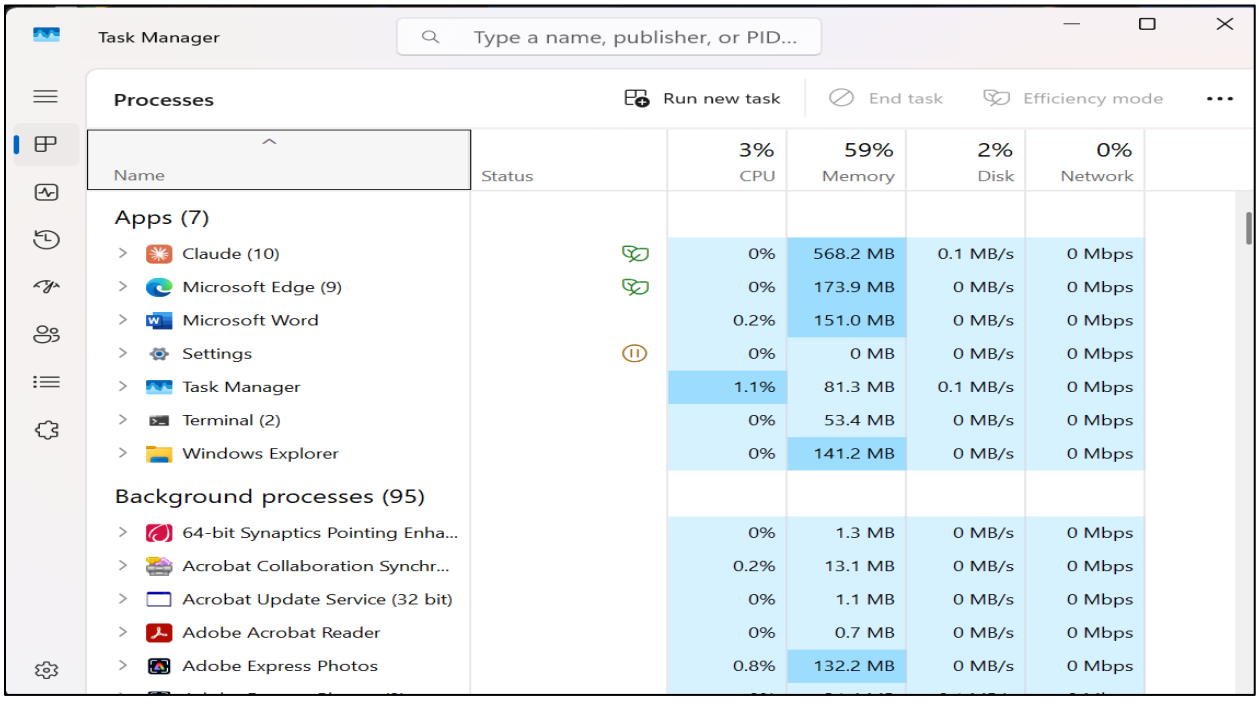
PURPOSE:

Opens Task View and virtual desktops.



- **Ctrl + Shift + Esc**

PURPOSE:



- **Alt + Tab**

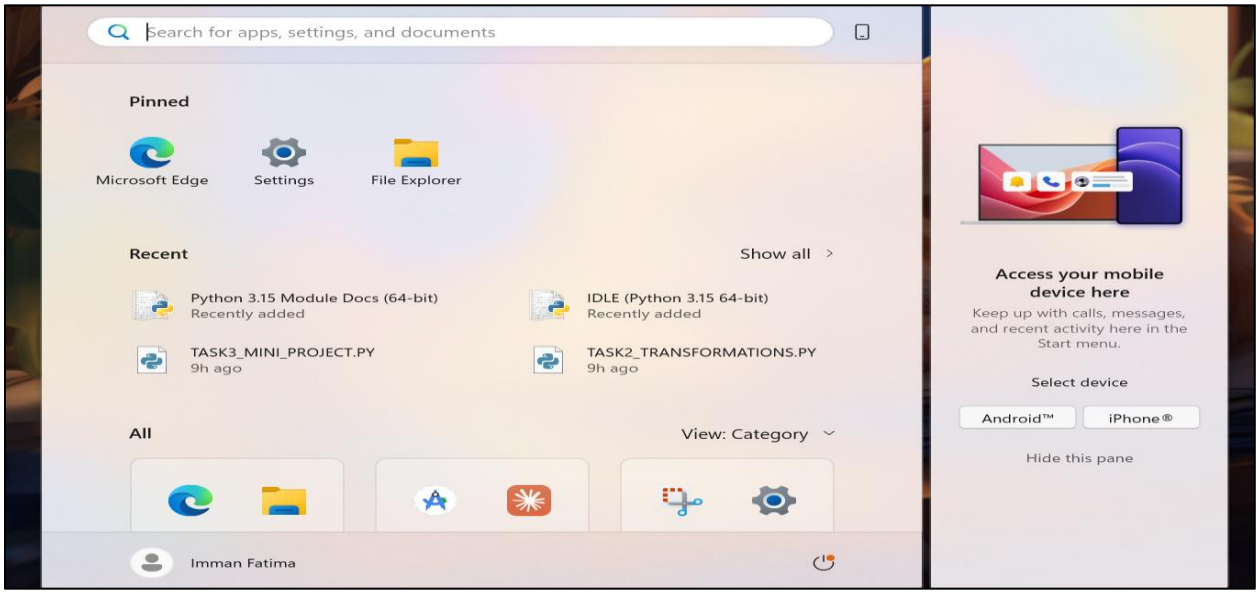
PURPOSE:

Switches between open windows.

- **Ctrl + Shift + Enter (in the Run box or Start search)**

PURPOSE:

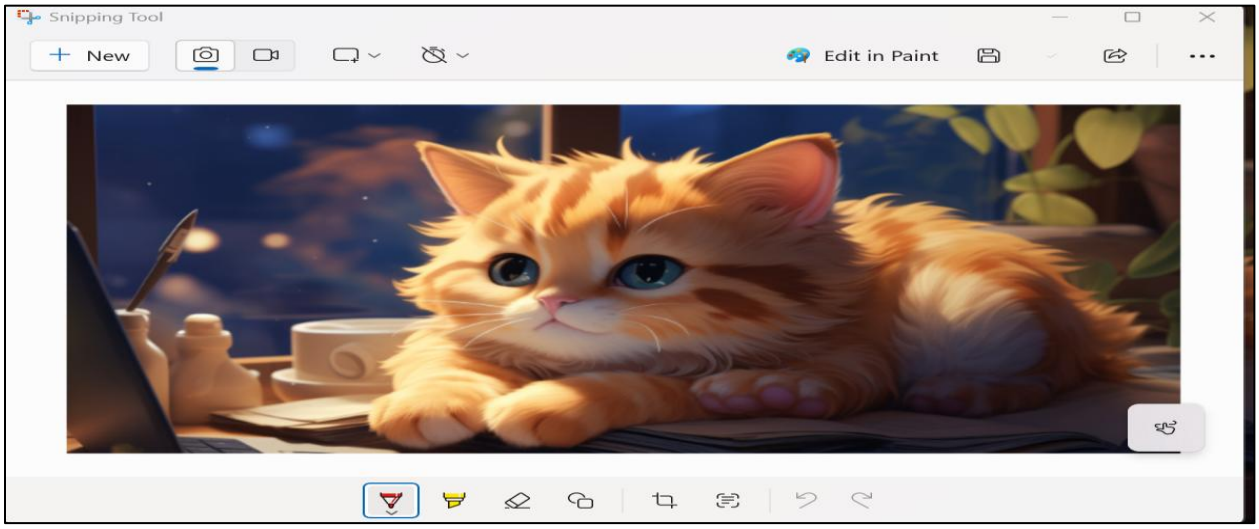
Opens an app as Administrator.



- **Win + Shift + S**

PURPOSE:

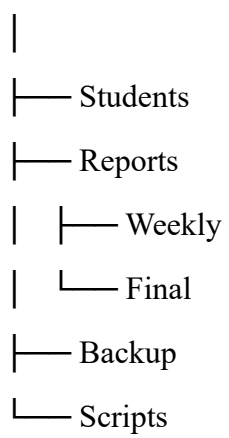
Opens the screenshot snipping tool.



Task 2 — File & Directory Management Using CMD

Open Command Prompt and create the following directory structure **without using File Explorer**:

OS_Lab_Assignment



After creating the structure:

1. **Display the current directory.**

```
C:\Users\123\OS_Lab_Assignment>cd
C:\Users\123\OS_Lab_Assignment
```

2. **Display the complete directory structure.**

```
C:\Users\123\OS_Lab_Assignment>mkdir Students

C:\Users\123\OS_Lab_Assignment>mkdir Reports

C:\Users\123\OS_Lab_Assignment>mkdir Reports\Weekly

C:\Users\123\OS_Lab_Assignment>mkdir Reports\Final

C:\Users\123\OS_Lab_Assignment>mkdir Backup

C:\Users\123\OS_Lab_Assignment>mkdir Scripts

C:\Users\123\OS_Lab_Assignment>tree /f
Folder PATH listing
Volume serial number is C260-4ABA
C:.
├───Backup
├───Reports
│   ├───Final
│   └───Weekly
├───Scripts
└───Students
```

3. **Create a text file named student_info.txt.**

```
C:\Users\123\OS_Lab_Assignment>echo Imman Fatima>student_info.txt
```

4. **Write your Name, Registration Number, Semester, and Section into the file using CMD.**

```
C:\Users\123\OS_Lab_Assignment>echo Name:Imman Fatima>>student_info.txt

C:\Users\123\OS_Lab_Assignment>echo Registration Number:242201037>>student_info.
txt

C:\Users\123\OS_Lab_Assignment>echo Semester:5th>>student_info.txt

C:\Users\123\OS_Lab_Assignment>echo Section:A>>student_info.txt
```

5. **Display the content of the file in CMD.**

```
C:\Users\123\OS_Lab_Assignment>type student_info.txt
Imman Fatima
Name:Imman Fatima
Registration Number:242201037
Semester:5th
Section:A
```

6. **Copy the file into the Backup folder.**

```
C:\Users\123\OS_Lab_Assignment>copy student_info.txt Backup
        1 file(s) copied.
```

7. **Rename the copied file to student_info_backup.txt.**

```
C:\Users\123\OS_Lab_Assignment>ren Backup\student_info.txt student_info_backup.txt

C:\Users\123\OS_Lab_Assignment>dir Backup
 Volume in drive C has no label.
 Volume Serial Number is C260-4ABA

 Directory of C:\Users\123\OS_Lab_Assignment\Backup

11/01/2026  03:51 PM    <DIR>          .
11/01/2026  03:26 PM    <DIR>          ..
11/01/2026  03:35 PM                89 student_info_backup.txt
               1 File(s)             89 bytes
               2 Dir(s)  412,077,711,360 bytes free
```

8. **Display all files and folders using CMD.**

```
C:\Users\123\OS_Lab_Assignment>dir/s
 Volume in drive C has no label.
 Volume Serial Number is C260-4ABA

 Directory of C:\Users\123\OS_Lab_Assignment

11/01/2026  03:26 PM    <DIR>          .
11/01/2026  03:17 PM    <DIR>          ..
11/01/2026  03:51 PM    <DIR>          Backup
11/01/2026  03:18 PM    <DIR>          Reports
11/01/2026  03:18 PM    <DIR>          Scripts
11/01/2026  03:17 PM    <DIR>          Students
11/01/2026  03:35 PM                89 student_info.txt
               1 File(s)             89 bytes

 Directory of C:\Users\123\OS_Lab_Assignment\Backup

11/01/2026  03:51 PM    <DIR>          .
11/01/2026  03:26 PM    <DIR>          ..
11/01/2026  03:35 PM                89 student_info_backup.txt
               1 File(s)             89 bytes

 Directory of C:\Users\123\OS_Lab_Assignment\Reports

11/01/2026  03:18 PM    <DIR>          .
11/01/2026  03:26 PM    <DIR>          ..
11/01/2026  03:18 PM    <DIR>          Final
11/01/2026  03:18 PM    <DIR>          Weekly
               0 File(s)              0 bytes

 Directory of C:\Users\123\OS_Lab_Assignment\Reports\Final

 Directory of C:\Users\123\OS_Lab_Assignment\Reports\Weekly

11/01/2026  03:18 PM    <DIR>          .
11/01/2026  03:18 PM    <DIR>          ..
               0 File(s)              0 bytes

 Directory of C:\Users\123\OS_Lab_Assignment\Scripts

11/01/2026  03:18 PM    <DIR>          .
11/01/2026  03:26 PM    <DIR>          ..
               0 File(s)              0 bytes

 Directory of C:\Users\123\OS_Lab_Assignment\Students

 Directory of C:\Users\123\OS_Lab_Assignment\Students

11/01/2026  03:17 PM    <DIR>          .
11/01/2026  03:26 PM    <DIR>          ..
               0 File(s)              0 bytes

     Total Files Listed:
               2 File(s)            178 bytes
              20 Dir(s)  412,075,528,192 bytes free
```

Task 3 — CMD System Investigation

Imagine that your supervisor asks you to quickly investigate the computer.

Using CMD, find and report:

- Computer name
- Current username
- Windows/IP configuration
- Current IP address
- Network connectivity status
- Running processes
- Current date and time

You must identify suitable CMD commands yourself.

For every command, write:

Command → Purpose → Output/Screenshot → Short Explanation

1. Computer Name

Command: hostname
Purpose: To find the name by which this computer is identified on the network.
Output:

```
C:\Users\123\Desktop\OS_Lab_Assignment>hostname
DESKTOP-17D2MKI
```

Explanation: The hostname command prints the computer's name. Administrators use it to confirm which machine they are working on, especially when managing multiple computers on a network.

2. Current Username

Command: whoami
Purpose: To identify the user account that is currently logged in and running the Command Prompt.
Output:

```
C:\Users\123\Desktop\OS_Lab_Assignment>whoami
desktop-17d2mki\123
```

Explanation: The whoami command displays the current user in the format computername\username. This helps an administrator verify which account is being used and what permissions it may have.

3. Windows / IP Configuration

Commands: ver and ipconfig /all
Purpose: To check the Windows version and view the complete network configuration of the computer.
Output:

```
C:\Users\123\Desktop\OS_Lab_Assignment>ver

Microsoft Windows [Version 10.0.26300.9550]
```

```
C:\Users\123\Desktop\OS_Lab_Assignment>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : DESKTOP-17D2MKI
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
   Physical Address. . . . . . . . . : B0-3C-DC-34-12-97
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
```

```
   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
   Physical Address. . . . . . . . . : B2-3C-DC-34-12-96
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Intel(R) Wi-Fi 6 AX201 160MHz
   Physical Address. . . . . . . . . : B0-3C-DC-34-12-96
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::856:951d:b5cf:2b2b%6(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.1.4(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Saturday, October 31, 2026 12:21:16 PM
   Lease Expires . . . . . . . . . . : Sunday, November 1, 2026 8:13:59 PM
   Default Gateway . . . . . . . . . : fe80::1%6
                                       192.168.1.1
   DHCP Server . . . . . . . . . . . : 192.168.1.1
   DHCPv6 IAID . . . . . . . . . . . : 61881564
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2D-9C-21-C0-B0-3C-DC-34-12-96
   DNS Servers . . . . . . . . . . . : 192.168.1.1
   Primary WINS Server . . . . . . . : 192.168.1.1
   Secondary WINS Server . . . . . . : 192.168.1.1
   NetBIOS over Tcpip. . . . . . . . : Enabled
```

Explanation: The ver command shows the installed Windows version. The ipconfig /all command displays detailed information for every network adapter, including the IP address, subnet mask, default gateway, DNS servers, DHCP status and MAC (physical) address.

4. Current IP Address

Command: ipconfig
Purpose: To find the IP address currently assigned to the computer.

Output:

```
C:\Users\123\Desktop\OS_Lab_Assignment>ipconfig

Windows IP Configuration


Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::856:951d:b5cf:2b2b%6
   IPv4 Address. . . . . . . . . . . : 192.168.1.4
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::1%6
                                       192.168.1.1
```

Explanation: The ipconfig command shows a summary of the network settings. The IPv4 Address line gives the computer's current IP address, which identifies it on the local network.

5. Network Connectivity Status

Command: ping 8.8.8.8
Purpose: To test whether the computer can communicate with another device over the network or internet.
Output:

```
C:\Users\123\Desktop\OS_Lab_Assignment>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=41ms TTL=115
Request timed out.
Reply from 8.8.8.8: bytes=32 time=42ms TTL=115
Reply from 8.8.8.8: bytes=32 time=46ms TTL=115

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 41ms, Maximum = 46ms, Average = 43ms
```

Explanation: The ping command sends small test packets to the given address (8.8.8.8 is a Google public DNS server) and waits for replies. Successful replies with a response time show that the connection is working. A “Request timed out” or “Destination host unreachable” message indicates a connectivity problem.

6. Running Processes

Command: tasklist
Purpose: To view the programs and services currently running on the computer.
Output:

```
svchost.exe                   4488 Services                   0      7,352 K
msedge.exe                    4240 Console                    34    200,892 K
msedge.exe                    9736 Console                    34     15,784 K
msedge.exe                   21132 Console                    34    105,432 K
msedge.exe                    2040 Console                    34     52,296 K
msedge.exe                    4736 Console                    34     23,200 K
msedge.exe                    7544 Console                    34      6,880 K
msedge.exe                    1160 Console                    34    128,064 K
msedge.exe                    7044 Console                    34     32,348 K
RuntimeBroker.exe             4600 Console                    34     27,812 K
ShellHost.exe                12420 Console                    34     47,660 K
msedge.exe                   19336 Console                    34     31,404 K
WINWORD.EXE                  10712 Console                    34    305,944 K
ai.exe                       20768 Console                    34     22,620 K
cmd.exe                       6760 Console                    34      6,568 K
conhost.exe                  11800 Console                    34      9,952 K
OpenConsole.exe               2560 Console                    34     14,332 K
WindowsTerminal.exe          18800 Console                    34     95,996 K
HPAudioAnalytics.exe         13512 Services                   0     15,340 K
FMToastNotification.exe       6152 Console                    34     11,964 K
Taskmgr.exe                  18188 Console                    34    156,592 K
taskhostw.exe                 6520 Console                    34     21,484 K
WindowsPackageManagerServ    19228 Console                    34     32,764 K
audiodg.exe                   7952 Services                   0     26,456 K
SearchProtocolHost.exe        3452 Services                   0     21,568 K
SearchFilterHost.exe         16888 Services                   0     12,592 K
SearchFilterHost.exe         18944 Services                   0     13,164 K
svchost.exe                  14548 Services                   0     33,388 K
svchost.exe                  14620 Services                   0     15,384 K
tasklist.exe                 10812 Console                    34     11,880 K
```

Explanation: The tasklist command lists all running processes along with their image name, process ID (PID), session name and memory usage. Administrators use it to monitor system activity and find programs that use too many resources.

7. Current Date and Time

Commands: date /t and time /t
Purpose: To display the current system date and time.
Output:

```
C:\Users\123\Desktop\OS_Lab_Assignment>date /t and time /t
Sat 10/31/2026
```

Explanation: The date /t command prints today's date and time /t prints the current time. The /t option only displays the value and does not ask for a new date or time to be entered, so the system settings stay unchanged.

Task 4 — Batch Script Automation

Your supervisor does not want to execute routine commands manually every day.

Create a batch file named system_check.bat.

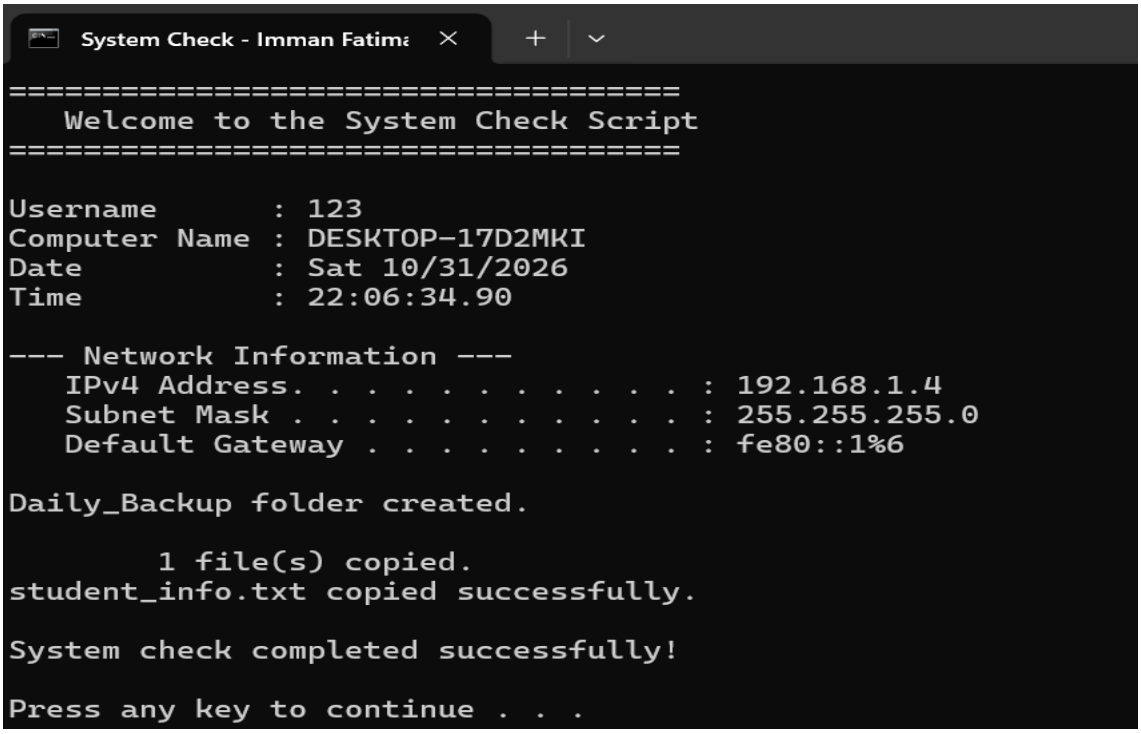
The script should automatically:

1. Display a welcome message.
2. Display the current username.
3. Display computer name.
4. Display current date and time.
5. Display basic IP/network information.
6. Create a folder named Daily_Backup if it does not already exist.
7. Copy student_info.txt into the backup folder.
8. Display a successful completion message.
9. Pause before closing.

Run the .bat file and provide screenshots showing its execution.

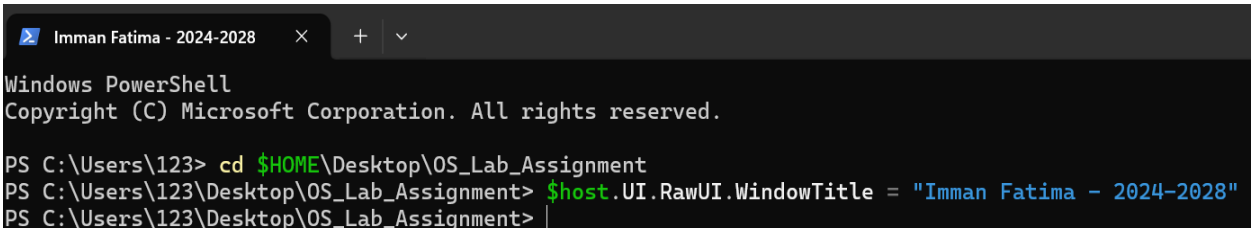
```
@echo off
title System Check - Imman Fatima
cd /d "%~dp0.."
echo ====================================
echo    Welcome to the System Check Script
echo ====================================
echo.
echo Username      : %USERNAME%
echo Computer Name : %COMPUTERNAME%
echo Date          : %DATE%
echo Time          : %TIME%
echo.
```

```
if not exist Daily_Backup (
    mkdir Daily_Backup
    echo Daily_Backup folder created.
) else (
    echo Daily_Backup folder already exists.
)
echo.
if exist student_info.txt (
    copy student_info.txt Daily_Backup\
    echo student_info.txt copied successfully.
    echo.
    echo System check completed successfully!
) else (
    echo ERROR: student_info.txt not found in %CD%
)
echo.
pause
```

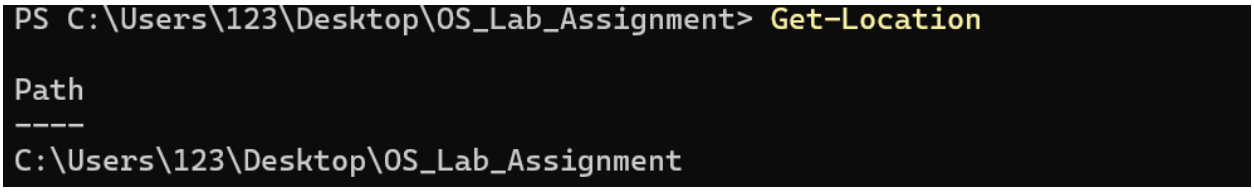


Task 5 — PowerShell Administration

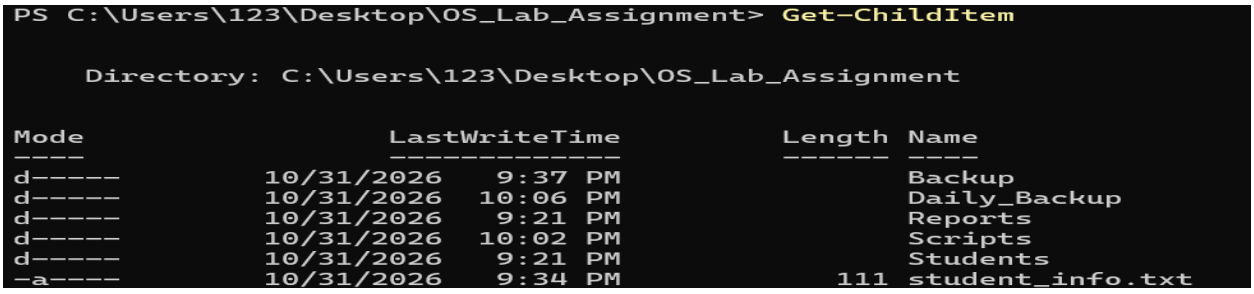
Now perform the following operations using PowerShell:



1. **Display the current working directory.**



2. **List files and folders.**



3. **Create a new folder named PowerShell_Task.**

```
PS C:\Users\123\Desktop\OS_Lab_Assignment> New-Item -ItemType Directory -Name PowerShell_Task


    Directory: C:\Users\123\Desktop\OS_Lab_Assignment


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        10/31/2026  10:13 PM                PowerShell_Task
```

4. **Create a text file inside it.**

```
PS C:\Users\123\Desktop\OS_Lab_Assignment> New-Item -ItemType File -Path .\PowerShell_Task\ps_info.txt


    Directory: C:\Users\123\Desktop\OS_Lab_Assignment\PowerShell_Task


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        10/31/2026  10:14 PM              0 ps_info.txt
```

5. **Write your Name and Registration Number into the file.**

```
PS C:\Users\123\Desktop\OS_Lab_Assignment> Set-Content .\PowerShell_Task\ps_info.txt "Name: Imman Fatima"
PS C:\Users\123\Desktop\OS_Lab_Assignment> Add-Content .\PowerShell_Task\ps_info.txt "Registration Number: 2024-2028"
```

6. **Read/display the contents of the file through PowerShell.**

```
PS C:\Users\123\Desktop\OS_Lab_Assignment> Get-Content .\PowerShell_Task\ps_info.txt
Name: Imman Fatima
Registration Number: 2024-2028
```

7. **Display currently running processes.**

```
PS C:\Users\123\Desktop\OS_Lab_Assignment> Get-Process | Select-Object -First 15

Handles  NPM(K)    PM(K)      WS(K)     CPU(s)      Id  SI ProcessName
-------  ------    -----      -----     ------      --  -- -----------
    172      10     2252      12140       9.66    2600  34 Adobe Crash Processor
    445      21     8232      30932       1.42    6732  34 AdobeCollabSync
    678      29    27016      48484      71.59    8864  34 AdobeCollabSync
    186      12     5040      15768                8076   0 AggregatorHost
    235      17    22452      22444       0.56   20768  34 ai
    357      19    12788      40372       1.38   23568  34 AppActions
    698      20     8132      24220                2952   0 AppHelperCap
    481      25    60720      49572       0.66   16208  34 ApplicationFrameHost
    383      20     5032      20888                2072   0 armsvc
    457      14    20704      26772       1.69   20536   0 audiodg
    496      21    22796      64388       0.48   16264  34 backgroundTaskHost
    370      26    96904      93520       7.45    7516  34 ChatGPT Classic
    197      12    10948      38572       0.23   14796  34 ChatGPT Classic
    381      19    15792      60668       2.86   18408  34 ChatGPT Classic
    749      25   126960      81292       2.64   19160  34 ChatGPT Classic
```

8. **Display basic system/computer information.**

```
PS C:\Users\123\Desktop\OS_Lab_Assignment> systeminfo | Select-Object -First 15

Host Name:                          DESKTOP-17D2MKI
OS Name:                            Microsoft Windows 11 Pro
OS Version:                         10.0.26300 N/A Build 26300
OS Manufacturer:                    Microsoft Corporation
OS Configuration:                   Standalone Workstation
OS Build Type:                      Multiprocessor Free
Registered Owner:                   123
Registered Organization:
Product ID:                         00330-80000-00000-AA534
Original Install Date:              4/1/2024, 11:30:06 AM
System Boot Time:                   10/4/2026, 11:36:37 AM
System Manufacturer:                HP
System Model:                       HP EliteBook 840 G8 Notebook PC
System Type:                        x64-based PC
```

9. **Copy your PowerShell-created file into the Backup folder.**

```
PS C:\Users\123\Desktop\OS_Lab_Assignment> Get-ChildItem .\Backup


    Directory: C:\Users\123\Desktop\OS_Lab_Assignment\Backup


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        10/31/2026  10:15 PM             52 ps_info.txt
-a----        10/31/2026   9:34 PM            111 student_info_backup.txt
```

Task 5: PowerShell Administration

Step	Command	Purpose	Explanation
1	Get-Location	Display the current working directory	Shows the full path of the folder where PowerShell is currently working.
2	Get-ChildItem	List files and folders	Lists all files and subfolders in the current directory.
3	New-Item -ItemType Directory -Name PowerShell_Task	Create a new folder	Creates a directory named PowerShell_Task in the current location.
4	New-Item -ItemType File -Path .\PowerShell_Task\ps_info.txt	Create a text file	Creates an empty text file named ps_info.txt inside the PowerShell_Task folder.
5	Set-Content and Add-Content	Write Name and Registration Number	Set-Content writes the first line into the file, and Add-Content appends a new line to it without erasing the old one.
6	Get-Content .\PowerShell_Task\ps_info.txt	Read the file	Displays the text stored in the file on the screen.

7	Get-Process	Display running processes	Lists all running processes with their ID, memory use and CPU time.
8	Get-ComputerInfo	Display basic system information	Shows details such as computer name, Windows version and installed memory.
9	Copy-Item .\PowerShell_Task\ps_info.txt .\Backup\	Copy the file to Backup	Makes a copy of ps_info.txt inside the Backup folder.

Task 6 — CMD vs PowerShell Comparison

Write a short comparison of CMD and PowerShell based on your practical experience.

Your comparison should discuss:

- Command style
- Ease of use
- Automation capability
- System administration capability
- Which one you would prefer for advanced Windows administration and why

Keep this section between **150–200 words**.

Task 6: CMD vs PowerShell Comparison

CMD and PowerShell are both command-line tools in Windows, but they differ in several ways. In terms of command style, CMD uses short, traditional commands such as dir, copy and ipconfig, while PowerShell uses a Verb-Noun style, for example Get-ChildItem and Copy-Item, which is longer but easier to understand. For ease of use, CMD felt simpler and faster for quick tasks like creating folders or checking an IP address, whereas PowerShell needed a little more learning. In automation, CMD is limited to batch files with basic syntax and weak error handling, while PowerShell supports variables, loops, functions and scripts, and its commands return objects that can be filtered and sorted through pipelines. For system administration, PowerShell can manage processes, services, the registry, users and even remote computers, while CMD mainly handles basic file and network tasks. Based on my practical work in this lab, I would prefer PowerShell for advanced Windows administration because it is more powerful, flexible and consistent, and it is Microsoft’s modern tool. CMD remains useful for simple tasks and older batch scripts.

Conclusion

Through this assignment I practiced using Windows keyboard shortcuts, CMD, batch scripting and PowerShell for basic system administration. I learned how to create and manage directories and files, investigate computer and network details, automate routine work with a batch file, and perform similar tasks in PowerShell. Comparing CMD and PowerShell showed me that command-line tools save time and make repeated administrative tasks easier and more reliable.